Youngju Na

aperxnt1@gmail.com · youngju-na.github.io · Google Scholar · LinkedIn · GitHub
Ph.D. student in Computer Science, KAIST

Research interests

Spatial intelligence for physical AI, spanning 3D scene representations, world models, and vision-language-action policies. My current research focuses on persistent world understanding and test-time adaptation for generalist robot policies.

Research & industry experience

Jan. 2027 – Apr. 2027
Applied Intuition – Sunnyvale, CA, USA
Incoming Ph.D. Research Intern, World-Action Foundation Model (Robotics)

Apr. 2025 – Oct. 2025
NAVER LABS, Spatial AI – South Korea
Research Intern | Advisor: Suyong Yeon
Research on scene-scale 3D reconstruction and neural rendering, including transparent surfaces.

Feb. 2023 – Present
Scalable Graphics, Vision, & Robotics Lab (SGVR Lab), KAIST – South Korea
Graduate Researcher | Advisor: Prof. Sung-Eui Yoon
Research in 3D vision, geometric scene representations, and robot learning.

Education

2025 – Present
Korea Advanced Institute of Science and Technology (KAIST)
Ph.D. in Computer Science

2023 – 2025
Korea Advanced Institute of Science and Technology (KAIST)
M.S. in Computer Science | GPA: 4.05/4.30

2017 – 2023
Chonnam National University
B.Eng. in IoT Artificial Intelligence and Industrial Engineering (double major)
Top Rank; GPA: 4.36/4.50, Major GPA: 4.46/4.50

Publications

* denotes equal contribution.

International conferences (selected first)

2026
GLINT: Modeling Scene-Scale Transparency via Gaussian Radiance Transport
Youngju Na, Jaeseong Yun, Soohyun Ryu, Hyunsu Kim, Sung-Eui Yoon, Suyong Yeon
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) **Oral; Best Paper Award Candidate** Project · Paper · Code

2026
Multi-Agent Coordination via Support-Preserving Distillation
Sangmin Lee, **Youngju Na**, Chanmi Lee, Sung-Eui Yoon
Conference on Neural Information Processing Systems (NeurIPS) Project · Paper

2026
MorphGS: Morphology-Adaptive Articulated 3D Motion Transfer from Videos
*Taeyeon Kim, ***Youngju Na**, Jumin Lee, Sebin Lee, Minhyuk Sung, Sung-Eui Yoon
European Conference on Computer Vision (ECCV) Project · Paper

2026
Visual-RRT: Finding Paths toward Visual-Goals via Differentiable Rendering
*Sebin Lee, *Jumin Lee, Taeyeon Kim, **Youngju Na**, Woobin Im, Sung-Eui Yoon
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) **Highlight** Project

2025
AegisRF: Adversarial Perturbations Guided with Sensitivity for Protecting Intellectual Property of Neural Radiance Fields
Woo Jae Kim, Kyu Beom Han, Yoonki Cho, **Youngju Na**, Junsik Jung, Sung-Eui Yoon
British Machine Vision Conference (BMVC) Project · Code

2025
Pose-Free 3D Gaussian Splatting via Shape-Ray Estimation
***Youngju Na**, *Taeyeon Kim, Jumin Lee, Woo Jae Kim, Kyu Beom Han, Sung-Eui Yoon
IEEE International Conference on Image Processing (ICIP) **Oral; Best Student Paper Award** Paper · Code

2024 **UFORecon: Generalizable Sparse-View Surface Reconstruction from Arbitrary and UnFavorable Sets**
Youngju Na, Woo Jae Kim, Kyu Beom Han, Suhyeon Ha, Sung-Eui Yoon
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Project · Paper · Code

2022 **HAZE-Net: High-Frequency Attentive Super-Resolved Gaze Estimation in Low-Resolution Face Images**
Jun-Seok Yun, **Youngju Na**, Hee Hyeon Kim, Hyung-Il Kim, Seok Bong Yoo
Asian Conference on Computer Vision (ACCV) Paper · Code

2022 **LatentGaze: Cross-Domain Gaze Estimation through Gaze-Aware Analytic Latent Code Manipulation**
Isack Lee, Jun-Seok Yun, Hee Hyeon Kim, **Youngju Na**, Seok Bong Yoo
Asian Conference on Computer Vision (ACCV) Paper · Code

International journal

2022 **Shared Knowledge Distillation for Robust Multi-Scale Super-Resolution Networks**
*__Youngju Na__, *Hee Hyeon Kim, Seok Bong Yoo
Electronics Letters 58(13) Paper

Domestic paper

2025 **Rectified Flow Distillation via 2nd Order Inversion for Consistent Text-to-3D Generation**
Sangmin Lee, Taeyeon Kim, **Youngju Na**, Jumin Lee, Sung-Eui Yoon
CKAIA

2023 **Articulated Motion Retargeting with 4D Gaussian Splatting**
*Taeyeon Kim, *__Youngju Na__, Sung-Eui Yoon
CKAIA

Selected honors & awards

2026 **Best Paper Award Candidate** (1.8% of accepted papers), CVPR

2025 **Best Student Paper Award** (0.16% of accepted papers), ICIP

2021 **1st Prize**, CNU Harvest Festival Symposium, AI/IoT Section, Chonnam National University

2021 **2nd Prize**, Silicon Valley Innovation & Startup Program, San Jose State University

Teaching experience

2024 Fall **Teaching Assistant**, Data Structure (CS206), KAIST

2024 Spring **Teaching Assistant**, Computer Graphics (CS380), KAIST

2023 Fall **Teaching Assistant**, Introduction to Programming (CS101), KAIST

2023 Spring **Teaching Assistant**, Introduction to Algorithms (CS300), KAIST

Academic service

Conference reviewer: ICLR 2026–2027; NeurIPS 2025–2026; ICCV 2025; CVPR 2026; AAAI 2025

Journal reviewer: IEEE Transactions on Multimedia (TMM); IEEE Transactions on Computational Imaging (TCI); International Journal of Computer Vision (IJCV)

Technical skills

Focus	3D reconstruction, neural rendering, robot learning, vision-language-action policies, world models
Programming	Python, C++, CUDA
ML / Vision	PyTorch, OpenCV, COLMAP, Blender / BlenderProc
Systems	Linux, Git, Docker, AWS
Languages	Korean (native), English (TOEFL iBT 102: R29/L25/S25/W23)